

Dirichlet-Multinomial Supplement

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Overview

- ▶ Simplex Review
- ▶ Dirichlet Distribution
- ▶ Categorical Distribution
- ▶ Multinomial Distribution
- ▶ Bayesian Inference
- ▶ Properties and Sampling Proofs

Review: The Simplex

The $(K - 1)$ -**dimensional simplex** Δ^{K-1} is the subspace of $\mathbb{R}^K$ on which K -component probability vectors live. That is,

$$\Delta^{K-1} = \left\{ (\pi_1, \dots, \pi_K) \in \mathbb{R}^K : \pi_k \geq 0, \sum_{k=1}^K \pi_k = 1 \right\}.$$

“($K - 1$)-dimensional” means that, once you have $(K - 1)$ components of the vector, there are no more free components.

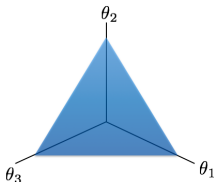


Figure: The two-dimensional simplex in $\mathbb{R}^3$.

The Dirichlet Distribution

The **K -component Dirichlet distribution** is a probability distribution over the $(K - 1)$ -dimensional simplex. It is a generalization of the Beta distribution.

It may be parameterized by a vector $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_K)$, $\alpha_k > 0$, of concentration parameters.

We then say

$$\boldsymbol{\pi} = (\pi_1, \dots, \pi_K) \sim \text{Dirichlet}(\boldsymbol{\alpha})$$

if

$$p(\pi_1, \dots, \pi_K) = \frac{\Gamma\left(\sum_{k=1}^K \alpha_k\right)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \pi_k^{\alpha_k - 1} \cdot \mathbb{1}[\boldsymbol{\pi} \in \Delta^{K-1}].$$

The Dirichlet Distribution

The Dirichlet may also be parameterized by a single concentration scalar $\alpha > 0$ and its mean vector $\boldsymbol{\nu} \in \Delta^{K-1}$, where $\boldsymbol{\alpha} = \alpha \boldsymbol{\nu}$.

To recover α and $\boldsymbol{\nu}$ from $\boldsymbol{\alpha}$, we normalize $\boldsymbol{\alpha}$ by the sum of its components (the 1-norm $\|\boldsymbol{\alpha}\|_1 = \sum_{k=1}^K |\alpha_k|$), getting

$$\alpha = \|\boldsymbol{\alpha}\|_1 \quad \boldsymbol{\nu} = \frac{\boldsymbol{\alpha}}{\|\boldsymbol{\alpha}\|_1} = \left(\frac{\alpha_1}{\sum_{k=1}^K \alpha_k}, \dots, \frac{\alpha_K}{\sum_{k=1}^K \alpha_k} \right).$$

Verify for yourself that $\boldsymbol{\nu} \in \Delta^{K-1}$.

The Dirichlet Distribution

When doing mathematical calculations with the Dirichlet density, one of the π_k – say, π_K – must be the “non-free” variable. That is,

$$p(\pi_1, \dots, \pi_K) = \frac{\Gamma\left(\sum_{k=1}^K \alpha_k\right)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \pi_k^{\alpha_k-1} \cdot \mathbb{1}[\boldsymbol{\pi} \in \Delta^{K-1}],$$

where π_K is a more compact way of writing $1 - \sum_{k=1}^{K-1} \pi_k$.

For the two-component Dirichlet,

$$p(\pi_1, \pi_2) = \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1) \Gamma(\alpha_2)} \pi_1^{\alpha_1-1} \pi_2^{\alpha_2-1} \cdot \mathbb{1}[\boldsymbol{\pi} \in \Delta^1].$$

Since there is only one free variable (pick π_1), this collapses to the univariate

$$p(\pi_1) = \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1) \Gamma(\alpha_2)} \pi_1^{\alpha_1-1} (1 - \pi_1)^{\alpha_2-1} \cdot \mathbb{1}[\pi_1 \in [0, 1]],$$

which is the Beta(α_1, α_2) distribution.

Dirichlet Mean

If $\boldsymbol{\pi} \sim \text{Dirichlet}(\boldsymbol{\alpha})$ with concentration $\alpha = \|\boldsymbol{\alpha}\|_1$ and mean vector $\boldsymbol{\nu}$, we have that

$$\begin{aligned}\mathbb{E}[\boldsymbol{\pi}] &= \boldsymbol{\nu} = \left(\frac{\alpha_1}{\|\boldsymbol{\alpha}\|_1}, \dots, \frac{\alpha_K}{\|\boldsymbol{\alpha}\|_1} \right) \\ &= \left(\frac{\alpha_1}{\sum_{k=1}^K \alpha_k}, \dots, \frac{\alpha_K}{\sum_{k=1}^K \alpha_k} \right) \\ &= (\mathbb{E}[\pi_1], \dots, \mathbb{E}[\pi_K]).\end{aligned}$$

$\boldsymbol{\nu}$ determines the location of the center of mass for the density, while α determines the direction and spread of the mass.

- ▶ $0 < \alpha < 1$: diffuse away from $\boldsymbol{\nu}$
- ▶ $\alpha = 1$: uniform
- ▶ $\alpha > 0$: concentrated toward $\boldsymbol{\nu}$

Examples

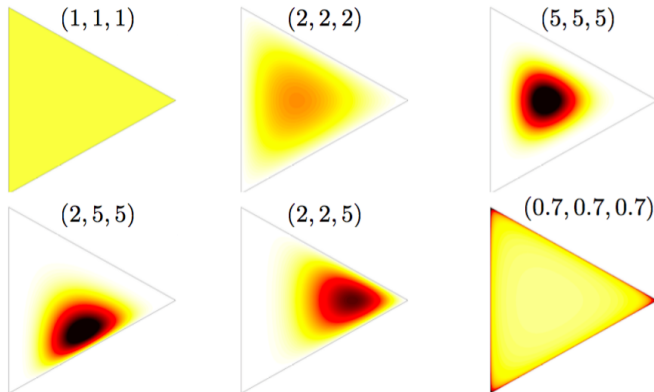


Figure: Top row: Mean at the centroid of the simplex, starting with uniform and getting more concentrated. Bottom row: Two concentrated and centered away from the centroid, and one centered at the centroid but diffuse to the vertices.

Examples

See DirichletDistributions.pdf (kindly provided by Mark Jacobson) in the Lecture 7 figures repository for non-heatmap example Dirichlet densities.

The Categorical Distribution

Suppose X can belong to one of K distinct categories, with the probability of belonging to category k equal to

$$\mathbb{P}(X = k) = \pi_k.$$

We say that X follows a **Categorical distribution**, parameterized by a K -component probability vector $\boldsymbol{\pi} \in \Delta^{K-1}$. We can alternatively express the PMF as

$$p(x) = \prod_{k=1}^K \pi_k^{\mathbb{1}[x=k]} \cdot \mathbb{1}[x \in \{1, \dots, K\}].$$

The Multinomial Distribution

Suppose you have n i.i.d. Categorical($\boldsymbol{\pi}$) draws and consider the distribution of their category counts $\mathbf{Y} = (Y_1, \dots, Y_K)$.

This vector of counts follows a **Multinomial distribution**, which is a multivariate generalization to the Binomial distribution.

Let $\mathbf{y} = (y_1, \dots, y_K)$. We say that

$$\mathbf{Y} = (Y_1, \dots, Y_K) \sim \text{Multinomial}(n, \boldsymbol{\pi})$$

if

$$\mathbb{P}(\mathbf{Y} = \mathbf{y} \mid \boldsymbol{\pi}) = \binom{n}{y_1 \cdots y_K} \prod_{k=1}^K \pi_k^{y_k} \cdot \mathbb{1} \left[\sum_{k=1}^K y_k = n \right],$$

where $\binom{n}{y_1 \cdots y_K}$ is the *multinomial coefficient* such that

$$\binom{n}{y_1 \cdots y_K} = \frac{n!}{\prod_{k=1}^K y_k!}.$$

The Multinomial Distribution

As with the Dirichlet distribution, one of the Y_k – say, Y_K – is not free. That is,

$$\mathbb{P}(\mathbf{Y} = \mathbf{y}) = \binom{n}{y_1 \cdots y_K} \prod_{k=1}^K \pi_k^{y_k} \cdot \mathbb{1} \left[\sum_{k=1}^K y_k = n \right],$$

where y_K is a compact way of writing $n - \sum_{k=1}^{K-1} y_k$.

For the two-component Multinomial,

$$\mathbb{P}(\mathbf{Y} = \mathbf{y}) = \binom{n}{y_1 \quad y_2} \pi_1^{y_1} \pi_2^{y_2} \cdot \mathbb{1} [y_1 + y_2 = n].$$

Pick y_1 as the free component; note that, since $\boldsymbol{\pi} \in \Delta^1$, we can also rewrite π_2 in terms of π_1 . So, this collapses to the univariate

$$\mathbb{P}(Y_1 = y_1) = \binom{n}{y_1} \pi_1^{y_1} (1 - \pi_1)^{n-y_1} \cdot \mathbb{1} [y_1 \in \{0, \dots, n\}],$$

which is the Binomial (n, π_1) distribution.

Posterior Updating

Suppose you have the following Bayesian model:

$$\begin{aligned}\mathbf{Y} \mid \boldsymbol{\pi} &\sim \text{Multinomial}(n, \boldsymbol{\pi}) \\ \boldsymbol{\pi} &\sim \text{Dirichlet}(\boldsymbol{\alpha}).\end{aligned}$$

Just as the Beta distribution is a conjugate prior to the Binomial likelihood, the Dirichlet is a conjugate prior to the Multinomial likelihood.

Posterior Updating

The posterior updating proceeds as follows:

$$\begin{aligned} p(\boldsymbol{\pi} \mid \mathbf{Y}) &\propto p(\mathbf{Y} \mid \boldsymbol{\pi}) \times p(\boldsymbol{\pi}) \\ &\propto \left(\binom{n}{y_1 \cdots y_K} \prod_{k=1}^K \pi_k^{y_k} \right) \times \left(\frac{\Gamma\left(\sum_{k=1}^K \alpha_k\right)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \pi_k^{\alpha_k-1} \right) \\ &\propto \left(\prod_{k=1}^K \pi_k^{y_k} \right) \times \left(\prod_{k=1}^K \pi_k^{\alpha_k-1} \right) \\ &\propto \prod_{k=1}^K \pi_k^{(\alpha_k + y_k) - 1}. \end{aligned}$$

The posterior then has the form

$$\boldsymbol{\pi} \mid \mathbf{Y} \sim \text{Dirichlet}(\boldsymbol{\alpha} + \mathbf{Y}).$$

Posterior Updating

Alternatively, we could deal with the Categorical likelihood, where

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Categorical}(\boldsymbol{\pi}).$$

We then have the following:

$$\begin{aligned} p(\mathbf{x}) &= \prod_{i=1}^n p(x_i) \\ &= \prod_{i=1}^n \prod_{k=1}^K \pi_k \mathbb{1}[x_i=k] \\ &= \prod_{k=1}^K \pi_k^{\sum_{i=1}^n \mathbb{1}[x_i=k]}. \end{aligned}$$

If we set $y_i = \sum_{k=1}^K \mathbb{1}[x_i = k]$, we see that

$$p(\mathbf{x}) = p(\mathbf{y}) = \prod_{k=1}^K \pi_k^{y_i}$$

recovers (up to proportionality) the Multinomial likelihood.

Interpreting Prior Parameters

Recall that, with the Beta (α_1, α_2) distribution, we identified

- ▶ α_1 : prior number of “successes”
- ▶ α_2 : prior number of “failures”
- ▶ $\alpha_1 + \alpha_2$: prior “sample size”

in the context of the Beta-Binomial scenario.

The interpretations are very similar for the Dirichlet(α) distribution! The main difference is that, instead of just two categories (success and failure), there are K categories. So, we have

- ▶ α_k : prior number of occurrences for category k
- ▶ $\sum_{k=1}^K \alpha_k$: prior sample size

Interpreting Prior Parameters

Recall the concentration-mean-vector parameterization $\text{Dirichlet}(\alpha, \boldsymbol{\nu})$, where

$$\alpha = \|\boldsymbol{\alpha}\|_1 = \sum_{k=1}^K \alpha_k \quad \text{and} \quad \boldsymbol{\nu} = \frac{\boldsymbol{\alpha}}{\|\boldsymbol{\alpha}\|_1} = \left(\frac{\alpha_1}{\sum_{k=1}^K \alpha_k}, \dots, \frac{\alpha_K}{\sum_{k=1}^K \alpha_k} \right).$$

The mean vector $\boldsymbol{\nu}$ can then be considered the prior category proportions.

The scalar concentration α is still the prior sample size, and its magnitude will reflect, in some sense, the confidence we have in our prior mean. (See [DirichletDistributions.pdf](#) again for an illustration.)

Summary

- ▶ Definition of the $(K - 1)$ -dimensional simplex
- ▶ Dirichlet density definition and illustrations
- ▶ Categorical and Multinomial densities
- ▶ Dirichlet-Multinomial conjugacy
- ▶ Interpreting the prior concentration parameters of the Dirichlet

Appendix: Generating a Beta Random Variate

Claim: If X_1 and X_2 are independent $\text{Gamma}(\alpha_1, \beta)$ and $\text{Gamma}(\alpha_2, \beta)$ RVs respectively, then $\frac{X_1}{X_1 + X_2} \sim \text{Beta}(\alpha_1, \alpha_2)$ (and $X_1 + X_2 \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta)$).

Proof: Start by writing the joint density of X_1 and X_2 ; by independence, we have

$$\begin{aligned} f_{X_1, X_2}(x_1, x_2) &= f_{X_1}(x_1) f_{X_2}(x_2) \\ &= \left(\frac{\beta^{\alpha_1}}{\Gamma(\alpha_1)} x_1^{\alpha_1-1} e^{-\beta x_1} \cdot \mathbb{1}[x_1 > 0] \right) \\ &\quad \times \left(\frac{\beta^{\alpha_2}}{\Gamma(\alpha_2)} x_2^{\alpha_2-1} e^{-\beta x_2} \cdot \mathbb{1}[x_2 > 0] \right) \\ &= \frac{\beta^{\alpha_1 + \alpha_2}}{\Gamma(\alpha_1) \Gamma(\alpha_2)} x_1^{\alpha_1-1} x_2^{\alpha_2-1} e^{-\beta(x_1 + x_2)} \cdot \mathbb{1}[x_1, x_2 > 0]. \end{aligned}$$

Appendix: Generating a Beta Random Variate

Let $Y = \frac{X_1}{X_1 + X_2}$ and $Z = X_1 + X_2$. We then solve for X_1 and X_2 in terms of Y and Z :

$$Y = \frac{X_1}{X_1 + X_2} = \frac{X_1}{Z} \Rightarrow X_1 = g_1(Y, Z) = YZ$$

$$Z = X_1 + X_2 = YZ + X_2 \Rightarrow X_2 = g_2(Y, Z) = Z(1 - Y).$$

From there, we set up the *Jacobian* (change-of-variable) matrix J :

$$\begin{aligned} J &= \begin{pmatrix} \frac{\partial X_1}{\partial Y} & \frac{\partial X_1}{\partial Z} \\ \frac{\partial X_2}{\partial Y} & \frac{\partial X_2}{\partial Z} \end{pmatrix} \\ &= \begin{pmatrix} Z & Y \\ -Z & 1 - Y \end{pmatrix}. \end{aligned}$$

We then take the absolute value of the determinant:

$$|\det(J)| = |Z(1 - Y) - Y(-Z)| = |Z - ZY + ZY| = |Z| = Z.$$

Appendix: Generating a Beta Random Variate

We then perform the following change of variables:

$$\begin{aligned} f_{Y,Z}(y, z) &= f_{X_1, X_2}(g_1(y, z), g_2(y, z)) \cdot |\det(J)| \\ &= \frac{\beta^{\alpha_1 + \alpha_2}}{\Gamma(\alpha_1) \Gamma(\alpha_2)} (yz)^{\alpha_1 - 1} ((1 - y)z)^{\alpha_2 - 1} e^{-\beta z} \\ &\quad \times \mathbb{1}[y \in (0, 1), z > 0] \cdot z \\ &= \frac{\beta^{\alpha_1 + \alpha_2}}{\Gamma(\alpha_1) \Gamma(\alpha_2)} y^{\alpha_1 - 1} (1 - y)^{\alpha_2 - 1} z^{\alpha_1 + \alpha_2 - 1} e^{-\beta z} \\ &\quad \times \mathbb{1}[y \in (0, 1)] \times \mathbb{1}[z > 0] \\ &= \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1) \Gamma(\alpha_2)} \cdot \frac{\beta^{\alpha_1 + \alpha_2}}{\Gamma(\alpha_1 + \alpha_2)} y^{\alpha_1 - 1} (1 - y)^{\alpha_2 - 1} z^{\alpha_1 + \alpha_2 - 1} e^{-\beta z} \\ &\quad \times \mathbb{1}[y \in (0, 1)] \times \mathbb{1}[z > 0] \\ &= \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1) \Gamma(\alpha_2)} y^{\alpha_1 - 1} (1 - y)^{\alpha_2 - 1} \times \mathbb{1}[y \in (0, 1)] \\ &\quad \times \frac{\beta^{\alpha_1 + \alpha_2}}{\Gamma(\alpha_1 + \alpha_2)} z^{\alpha_1 + \alpha_2 - 1} e^{-\beta z} \times \mathbb{1}[z > 0]. \end{aligned}$$

Appendix: Generating a Beta Random Variate

So, we have that $f_{Y,Z}(y, z) = f_Y(y)f_Z(z)$, where

$$f_Y(y) = \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y^{\alpha_1-1} (1-y)^{\alpha_2-1} \times \mathbb{1}[y \in (0, 1)]$$

and

$$f_Z(z) = \frac{\beta^{\alpha_1+\alpha_2}}{\Gamma(\alpha_1 + \alpha_2)} z^{\alpha_1+\alpha_2-1} e^{-\beta z} \times \mathbb{1}[z > 0].$$

Therefore, when X_1 and X_2 are independent Gammas with equal rate parameter,

$$Y = \frac{X_1}{X_1 + X_2} \sim \text{Beta}(\alpha_1, \alpha_2) \quad Z = X_1 + X_2 \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta).$$

Further, Y and Z are themselves independent. □

Exercise: Define $Y = \frac{X_2}{X_1 + X_2}$ and show that the parameters of the resulting Beta switch.

Appendix: Generating a Dirichlet Random Vector

Claim: If $X_1, \dots, X_K$ are independent Gamma (α_k, β) RVs respectively, then

$$\left(\frac{X_1}{\sum_{k=1}^K X_k}, \dots, \frac{X_K}{\sum_{k=1}^K X_k} \right) \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_K).$$

Proof: Write the joint density of the X_k :

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}) &= \prod_{k=1}^K f_{X_i}(x_i) \\ &= \prod_{k=1}^K \frac{\beta^{\alpha_k}}{\Gamma(\alpha_k)} x_k^{\alpha_k-1} e^{-\beta x_k} \times \mathbb{1}[x_k > 0] \\ &= \frac{\beta^{\sum_{k=1}^K \alpha_k}}{\prod_{k=1}^K \Gamma(\alpha_k)} \left(\prod_{k=1}^K x_k^{\alpha_k-1} \right) e^{-\beta \sum_{k=1}^K x_k} \times \mathbb{1}[x_1, \dots, x_k > 0]. \end{aligned}$$

Appendix: Generating a Dirichlet Random Vector

For $k = 1, \dots, K - 1$, let $Y_k = \frac{X_k}{\sum_{k=1}^K X_k}$. Let $Z = \sum_{k=1}^K X_k$. We then solve for the X_k in terms of the Y_k and Z :

$$Y_k = \frac{X_k}{\sum_{k=1}^K X_k} = \frac{X_k}{Z} \Rightarrow X_k = g_k(Y_k, Z) = Y_k Z \quad \text{for } k = 1, \dots, K-1$$

$$Z = \sum_{k=1}^{K-1} Y_k Z + X_K \Rightarrow X_K = g_K(\mathbf{Y}, Z) = Z \left(1 - \sum_{k=1}^{K-1} Y_k \right).$$

When we set up the $K \times K$ Jacobian, note that, for $k \in \{1, \dots, K - 1\}$,

$$\frac{\partial X_j}{\partial Y_k} = 0 \quad \text{for } j \neq k \quad \frac{\partial X_k}{\partial Y_k} = Z \quad \frac{\partial X_k}{\partial Z} = Y_k$$

$$\frac{\partial X_K}{\partial Y_k} = -Z \quad \frac{\partial X_K}{\partial Z} = 1 - \sum_{k=1}^{K-1} Y_k.$$

Appendix: Generating a Dirichlet Random Vector

We set up the Jacobian:

$$\begin{aligned} J &= \begin{pmatrix} \frac{\partial X_1}{\partial Y_1} & \cdots & \frac{\partial X_1}{\partial Y_{K-1}} & \frac{\partial X_1}{\partial Z} \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial X_{K-1}}{\partial Y_1} & \cdots & \frac{\partial X_{K-1}}{\partial Y_{K-1}} & \frac{\partial X_{K-1}}{\partial Z} \\ \frac{\partial X_K}{\partial Y_1} & \cdots & \frac{\partial X_K}{\partial Y_{K-1}} & \frac{\partial X_K}{\partial Z} \end{pmatrix} \\ &= \begin{pmatrix} Z & \cdots & 0 & Y_1 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & Z & Y_{K-1} \\ -Z & \cdots & -Z & 1 - \sum_{k=1}^{K-1} Y_k \end{pmatrix}. \end{aligned}$$

We then have that $|\det(J)| = Z^{K-1}$.

Appendix: Generating a Dirichlet Random Vector

We then perform the following change of variables, setting

$$y_K = 1 - \sum_{k=1}^K y_k:$$

$$\begin{aligned} f_{\mathbf{Y},Z}(\mathbf{y}, z) &= f_{\mathbf{X}}(g_1(\mathbf{y}, z), \dots, g_K(\mathbf{y}, z)) \cdot |\det(J)| \\ &= \frac{\beta^{\sum_{k=1}^K \alpha_k}}{\prod_{k=1}^K \Gamma(\alpha_k)} \left(\prod_{k=1}^K (y_k z)^{\alpha_k - 1} \right) e^{-\beta z} \\ &\quad \times \mathbb{1}[\mathbf{y} \in \Delta^{K-1}, z > 0] \cdot z^{K-1} \\ &= \frac{\beta^{\sum_{k=1}^K \alpha_k}}{\prod_{k=1}^K \Gamma(\alpha_k)} \left(\prod_{k=1}^K y_k^{\alpha_k - 1} \right) z^{\sum_{k=1}^K \alpha_k - 1} e^{-\beta z} \\ &\quad \times \mathbb{1}[\mathbf{y} \in \Delta^{K-1}] \times \mathbb{1}[z > 0] \\ &= \frac{\Gamma\left(\sum_{k=1}^K \alpha_k\right)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K y_k^{\alpha_k - 1} \times \mathbb{1}[\mathbf{y} \in \Delta^{K-1}] \\ &\quad \times \frac{\beta^{\sum_{k=1}^K \alpha_k}}{\Gamma\left(\sum_{k=1}^K \alpha_k\right)} z^{\sum_{k=1}^K \alpha_k - 1} e^{-\beta z} \times \mathbb{1}[z > 0]. \end{aligned}$$

Appendix: Generating a Dirichlet Random Vector

So, we have that $f_{\mathbf{Y},Z}(\mathbf{y}, z) = f_{\mathbf{Y}}(\mathbf{y})f_Z(z)$, where

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{\Gamma\left(\sum_{k=1}^K \alpha_k\right)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K y_k^{\alpha_k-1} \times \mathbb{1}[\mathbf{y} \in \Delta^{K-1}]$$

and

$$f_Z(z) = \frac{\beta \sum_{k=1}^K \alpha_k}{\Gamma\left(\sum_{k=1}^K \alpha_k\right)} z^{\sum_{k=1}^K \alpha_k-1} e^{-\beta z} \times \mathbb{1}[z > 0].$$

Therefore, when $X_1, \dots, X_K$ are independent Gammas with equal rate parameter,

$$\left(\frac{X_1}{\sum_{k=1}^K X_k}, \dots, \frac{X_K}{\sum_{k=1}^K X_k} \right) \sim \text{Dirichlet}(\boldsymbol{\alpha})$$
$$\sum_{k=1}^K X_k \sim \text{Gamma}\left(\sum_{k=1}^K \alpha_k, \beta\right). \quad \square$$

Appendix: Marginal Properties

Claim: The multivariate joint marginals of a Multinomial distribution are Multinomial, while the univariate marginals of a Multinomial distribution are Binomial.

Claim: The multivariate joint marginals of a Dirichlet distribution are Dirichlet, while the univariate marginals of a Dirichlet distribution are Beta.

Proof: Left as an exercise!